

Managing Climate Risk: Sovereign and Corporate Debt Markets

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Abstract

Physical climate risks such as droughts, floods, and extreme heat are increasingly recognized as systematic risk factors for the real economy and financial markets, yet we know comparatively little about whether and how public policy can reshape their pricing.

Keywords: Physical climate risks, asset pricing, public policy

1. Introduction

2. Model

2.1. Economic Environment, Climate Risk, and Adaptation

Time is discrete, $t = 0, 1, 2, \dots$. Uncertainty is generated by a climate risk factor X_t and other aggregate shocks. Government adaptation investment is denoted by

$G_t \geq 0$, which can be interpreted as the flow of public spending on climate adaptation (e.g., large-scale water projects, flood defenses). For now, we treat G_t as exogenous and focus on its asset-pricing implications.

Climate risk dynamics. We model the climate factor X_t as a (possibly persistent) shock that captures the intensity of physical climate risk (e.g., drought risk) at time t . Its dynamics are given by

$$X_{t+1} = (1 - \phi(G_t))X_t + \sigma(G_t)\varepsilon_{t+1}, \quad (1)$$

where ε_{t+1} is an i.i.d. innovation with mean zero and unit variance, $\phi(G_t) \in [0, 1]$ measures the mean-reversion of the climate shock, and $\sigma(G_t) > 0$ is the conditional volatility of X_{t+1} .

We assume:

$$\phi'(G) > 0, \quad \sigma'(G) < 0. \quad (2)$$

That is, a higher level of adaptation investment G increases the mean-reversion of climate shocks and reduces their conditional variance. Under (1)–(2), G reduces both the persistence and volatility of climate risk.

Aggregate output and tax base. Aggregate output next period is given by

$$Y_{t+1} = \bar{Y} - d_X X_{t+1} + u_{t+1}, \quad (3)$$

where $d_X > 0$ captures the adverse impact of climate risk on output and u_{t+1} is an i.i.d. disturbance. The government collects proportional taxes

$$T_{t+1} = \tau Y_{t+1}, \quad \tau \in (0, 1). \quad (4)$$

Stochastic discount factor. Asset prices are determined by a stochastic discount factor (SDF) M_{t+1} . We do not commit to a particular preference specification but assume that climate risk is *bad* in the sense that

$$\text{Cov}_t(M_{t+1}, X_{t+1}) > 0. \quad (5)$$

Intuitively, high realizations of X_{t+1} (more severe climate risk) are associated with high marginal utility and thus high M_{t+1} .

2.2. Government Sector and Sovereign Debt Pricing

The government issues one-period defaultable bonds. Let B_t denote the stock of outstanding government debt at time t , and let r_t^f be the risk-free short rate. The government budget constraint is

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \quad (6)$$

where r_t^g is the required return on sovereign bonds, G_t is adaptation spending, and I_t collects other government expenditures.

Default intensity. We model sovereign default in reduced form. Let $D_{t+1}^g \in \{0, 1\}$ be the indicator that the government defaults between t and $t + 1$. The (physical) default intensity is

$$\lambda_t^g = \Lambda^g(X_t, B_t, G_t), \quad (7)$$

where $\Lambda^g(\cdot)$ is increasing in climate risk and debt, but decreasing in adaptation:

$$\frac{\partial \Lambda^g}{\partial X} > 0, \quad \frac{\partial \Lambda^g}{\partial B} > 0, \quad \frac{\partial \Lambda^g}{\partial G} < 0. \quad (8)$$

These assumptions capture the idea that severe climate risk and high debt raise sovereign default risk, while greater adaptation investment improves macroeconomic resilience and reduces default intensity.

Sovereign bond pricing and spreads. Consider a one-period sovereign bond with face value one and price P_t^g . Its payoff is $(1 - D_{t+1}^g)$ in units of the numeraire. The no-arbitrage pricing equation is

$$P_t^g = \mathbb{E}_t \left[M_{t+1} (1 - D_{t+1}^g) \right]. \quad (9)$$

Define the gross required return

$$1 + r_t^g := \frac{1}{P_t^g}. \quad (10)$$

Let $P_t^f = \mathbb{E}_t[M_{t+1}]$ denote the price of a default-free one-period bond; its return satisfies $1 + r_t^f = 1/P_t^f$. The *sovereign spread* is defined as

$$s_t^g := r_t^g - r_t^f. \quad (11)$$

Under a standard linearization of (9), the sovereign spread can be decomposed into an expected-loss component and a risk-premium component linked to climate risk.¹

Proposition 1 (Decomposition of sovereign spreads) *Under mild regularity conditions and a local linearization around the mean, the sovereign spread can be written as*

$$s_t^g \approx \underbrace{\lambda_t^{g,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_X^g(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (12)$$

where $\lambda_t^{g,\mathbb{P}}$ is the physical default intensity, $\beta_X^g(G_t)$ is the sovereign bond's exposure (beta) to the climate factor X_{t+1} , and $\lambda_X(G_t)$ is the market price of climate risk. Both $\beta_X^g(G)$ and $\lambda_X(G)$ are functions of the level of adaptation G through the climate dynamics (1)–(2) and the SDF covariance (5).

We next characterize how an increase in adaptation G_t affects the sovereign spread.

Proposition 2 (Climate adaptation and sovereign financing costs) *Suppose that:*

¹The exact functional form is not central for our comparative statics. A formal derivation is provided in the Appendix XX.

1. Λ^g satisfies the monotonicity conditions (8);
2. G affects the climate dynamics through (1)–(2);
3. the SDF satisfies (5).

Then, holding (X_t, B_t) fixed, the sovereign spread s_t^g is decreasing in G_t :

$$\frac{\partial s_t^g}{\partial G_t} < 0. \quad (13)$$

The total effect can be decomposed into:

$$\frac{\partial s_t^g}{\partial G_t} = \underbrace{\frac{\partial \lambda_t^{g, \mathbb{P}}}{\partial G_t}}_{<0, \text{ expected loss effect}} + \underbrace{\frac{\partial \beta_X^g(G_t)}{\partial G_t} \lambda_X(G_t) + \beta_X^g(G_t) \frac{\partial \lambda_X(G_t)}{\partial G_t}}_{\leq 0, \text{ risk premium effect}}.$$

Proof sketch. The expected-loss effect follows directly from (7)–(8). The risk-premium effect arises because G reduces both the variance of X_{t+1} and its persistence, thereby lowering the sovereign bond’s exposure $\beta_X^g(G)$, and because the market price of climate risk $\lambda_X(G)$ is decreasing in the volatility of X_{t+1} given (5). A formal derivation is provided in Appendix XX. ■

Corollary 1 (Cross-sectional heterogeneity in sovereign spread relief) *The impact of adaptation investment G_t on sovereign spreads is stronger when:*

1. the economy is more climate-exposed (higher d_X in (3)), and/or
2. the government debt level B_t is higher (so that a given reduction in default intensity generates a larger change in expected loss).

2.3. Corporate Sector and Credit Spreads

We now introduce a continuum of firms indexed by i . Each firm issues defaultable corporate bonds and is exposed to both climate risk and sovereign risk.

Corporate cash flows and climate exposure. Firm i 's cash flow next period is

$$CF_{i,t+1} = \bar{C}F_i - \theta_i X_{t+1} - \eta_i \lambda_t^g + \epsilon_{i,t+1}, \quad (14)$$

where $\theta_i \geq 0$ measures the firm's technological exposure to climate risk (e.g., water-intensive production), $\eta_i \geq 0$ captures its exposure to sovereign risk (e.g., dependence on fiscal transfers or regulated tariffs), and $\epsilon_{i,t+1}$ is an idiosyncratic shock.

Corporate default intensity. Firm i 's (physical) default intensity is modeled as

$$\lambda_{i,t}^c = \Lambda^c(X_t, G_t; Z_i), \quad (15)$$

where Z_i collects firm-level characteristics. We assume

$$\frac{\partial \Lambda^c}{\partial X} > 0, \quad \frac{\partial \Lambda^c}{\partial G} < 0, \quad (16)$$

so that worse climate conditions raise default risk, while adaptation investment attenuates it.

Corporate bond pricing and spreads. Consider a one-period zero-coupon bond issued by firm i with price $P_{i,t}^c$ and payoff $(1 - D_{i,t+1}^c)$. The price satisfies

$$P_{i,t}^c = \mathbb{E}_t \left[M_{t+1} (1 - D_{i,t+1}^c) \right], \quad (17)$$

where $D_{i,t+1}^c$ is the default indicator. The corporate bond return is $1 + r_{i,t}^c := 1/P_{i,t}^c$, and the corporate credit spread is

$$s_{i,t}^c := r_{i,t}^c - r_t^f. \quad (18)$$

Parallel to Proposition 1 and Proposition 2, we obtain:

Proposition 3 (Decomposition of corporate credit spreads) *Under a local linearization, the corporate credit spread can be written as*

$$s_{i,t}^c \approx \underbrace{\lambda_{i,t}^{c,\mathbb{P}}}_{\text{expected loss}} + \underbrace{\beta_{i,X}^c(G_t) \lambda_X(G_t)}_{\text{climate risk premium}} + (\text{other risk premia}), \quad (19)$$

where $\lambda_{i,t}^{c,\mathbb{P}}$ is the firm's physical default intensity, $\beta_{i,X}^c(G_t)$ is the firm's bond exposure to climate risk, and $\lambda_X(G_t)$ is the same market price of climate risk as in (12).

Proposition 4 (Climate adaptation and corporate financing costs) *If (16) holds and G affects the climate dynamics as in (1)–(2), then, holding (X_t, Z_i) fixed, the corporate credit spread $s_{i,t}^c$ is decreasing in adaptation investment:*

$$\frac{\partial s_{i,t}^c}{\partial G_t} < 0. \quad (20)$$

The reduction in spreads is stronger for firms with larger climate-exposure coefficients θ_i and sovereign-risk loadings η_i in (14), as summarized in Corollary 2.

Corollary 2 (Cross-sectional predictions for corporate spreads) *For a given change in adaptation investment G_t , the decline in credit spreads $s_{i,t}^c$ is predicted to be:*

1. *larger for firms with higher θ_i (more climate-sensitive cash flows, e.g., water-intensive industries); and*
2. *larger for firms with higher η_i (stronger dependence on sovereign risk, e.g., state-owned enterprises or regulated utilities).*

These cross-sectional predictions provide testable implications for the empirical analysis.

2.4. Joint Government–Corporate Financing Conditions

Finally, define the average sovereign spread $\bar{s}_t^g := \mathbb{E}[s_t^g]$ and an aggregate corporate spread $\bar{s}_t^c := \mathbb{E}_i[s_{i,t}^c]$. Combining Propositions 2 and 4, we obtain:

Proposition 5 (Adaptation investment and aggregate financing costs) *Under the above assumptions, an increase in adaptation investment G_t reduces both sovereign and corporate spreads at the aggregate level:*

$$\frac{\partial \bar{s}_t^g}{\partial G_t} < 0, \quad \frac{\partial \bar{s}_t^c}{\partial G_t} < 0. \quad (21)$$

Moreover, the joint reduction in public and private financing costs can offset part of the contemporaneous fiscal cost of adaptation G_t in the government budget constraint (6).

The next section connects these model-based predictions to sovereign and corporate bond data, with a focus on the effects of large-scale adaptation projects (such as major water transfer schemes) on financing conditions in both the public and corporate sectors.

3. Empirical Analysis

A. Proofs

A.1. Proof of Proposition 1: Decomposition of sovereign spreads

Consider the one-period defaultable sovereign bond with price

$$P_t^g = \mathbb{E}_t[M_{t+1}(1 - D_{t+1}^g)], \quad (\text{A.1})$$

and the default-free bond price

$$P_t^f = \mathbb{E}_t[M_{t+1}], \quad 1 + r_t^f = \frac{1}{P_t^f}. \quad (\text{A.2})$$

Define the physical default probability (intensity over one period)

$$\lambda_t^{g,\mathbb{P}} := \mathbb{E}_t[D_{t+1}^g]. \quad (\text{A.3})$$

Step 1: Price wedge between defaultable and risk-free bonds. Rewrite (A.1) as

$$\begin{aligned} P_t^g &= \mathbb{E}_t[M_{t+1}] - \mathbb{E}_t[M_{t+1}D_{t+1}^g] \\ &= P_t^f - \left(\mathbb{E}_t[M_{t+1}]\mathbb{E}_t[D_{t+1}^g] + \text{Cov}_t(M_{t+1}, D_{t+1}^g) \right) \\ &= P_t^f(1 - \lambda_t^{g,\mathbb{P}}) - \text{Cov}_t(M_{t+1}, D_{t+1}^g). \end{aligned} \quad (\text{A.4})$$

Hence,

$$\frac{P_t^g}{P_t^f} = 1 - \lambda_t^{g,\mathbb{P}} - \frac{\text{Cov}_t(M_{t+1}, D_{t+1}^g)}{P_t^f}. \quad (\text{A.5})$$

Step 2: Linearize the yield (return) spread. By definition,

$$1 + r_t^g = \frac{1}{P_t^g}, \quad s_t^g := r_t^g - r_t^f. \quad (\text{A.6})$$

Note that

$$\frac{1 + r_t^g}{1 + r_t^f} = \frac{P_t^f}{P_t^g} = \left(\frac{P_t^g}{P_t^f} \right)^{-1}. \quad (\text{A.7})$$

Let

$$x_t := \lambda_t^{g, \mathbb{P}} + \frac{\text{Cov}_t(M_{t+1}, D_{t+1}^g)}{P_t^f}.$$

Then (A.5) implies $\frac{P_t^g}{P_t^f} = 1 - x_t$. Under a local approximation around small x_t (mild default risk and small pricing wedge), use $(1 - x)^{-1} \approx 1 + x$:

$$\frac{1 + r_t^g}{1 + r_t^f} = \frac{P_t^f}{P_t^g} \approx 1 + x_t. \quad (\text{A.8})$$

Therefore,

$$\begin{aligned} s_t^g &= r_t^g - r_t^f = (1 + r_t^f) \left(\frac{1 + r_t^g}{1 + r_t^f} - 1 \right) \\ &\approx (1 + r_t^f) \left(\lambda_t^{g, \mathbb{P}} + \frac{\text{Cov}_t(M_{t+1}, D_{t+1}^g)}{P_t^f} \right). \end{aligned} \quad (\text{A.9})$$

Since $1 + r_t^f = 1/P_t^f$ and the one-period risk-free rate is typically small, a common simplification is to absorb $(1 + r_t^f)$ into the approximation error and write

$$s_t^g \approx \lambda_t^{g, \mathbb{P}} + \frac{\text{Cov}_t(M_{t+1}, D_{t+1}^g)}{P_t^f}. \quad (\text{A.10})$$

The first term $\lambda_t^{g,\mathbb{P}}$ is the expected-loss component.

Step 3: Factor-risk (climate) decomposition of the covariance term. Let $\tilde{X}_{t+1} := X_{t+1} - \mathbb{E}_t[X_{t+1}]$ denote the innovation in the climate factor. Following, Cochrane (2005), project the default indicator on the span of $\tilde{X}_{t+1}$ and other systematic innovations $\tilde{Z}_{t+1}$:

$$D_{t+1}^g = b_X^g(G_t)\tilde{X}_{t+1} + (b_Z^g)^\top \tilde{Z}_{t+1} + \varepsilon_{t+1}^g, \quad \text{Cov}_t(\varepsilon_{t+1}^g, \tilde{X}_{t+1}) = 0. \quad (\text{A.11})$$

Then

$$\text{Cov}_t(M_{t+1}, D_{t+1}^g) = b_X^g(G_t)\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1}) + (b_Z^g)^\top \text{Cov}_t(M_{t+1}, \tilde{Z}_{t+1}) + \text{Cov}_t(M_{t+1}, \varepsilon_{t+1}^g). \quad (\text{A.12})$$

Define the (conditional) market price of climate risk as

$$\lambda_X(G_t) := -\frac{\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1})}{\mathbb{E}_t[M_{t+1}]} = -\frac{\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1})}{P_t^f}. \quad (\text{A.13})$$

Moreover, define the sovereign bond's (climate) beta (up to the same first-order scaling used in the price-to-yield linearization in Step 2) as

$$\beta_X^g(G_t) := -(1 + r_t^f) b_X^g(G_t). \quad (\text{A.14})$$

Substituting (A.12) into (A.9) and using (A.13)–(A.14) yields

$$s_t^g \approx \lambda_t^{g,\mathbb{P}} + \beta_X^g(G_t)\lambda_X(G_t) + (\text{other risk premia}), \quad (\text{A.15})$$

which corresponds to equation (12) in the main text.

A.2. Proof of Proposition 2: Climate adaptation and sovereign financing costs

We work under the same local linearization used in Proposition 1 and Appendix A.1.

Holding (X_t, B_t) fixed, Proposition 1 implies

$$s_t^g \approx \lambda_t^{g,\mathbb{P}} + \beta_X^g(G_t)\lambda_X(G_t) + (\text{other risk premia}).$$

Throughout, we treat “other risk premia” as orthogonal to the comparative statics in G_t (e.g., either constant in G_t conditional on (X_t, B_t) , or weakly decreasing in G_t). This is without loss for the sign result, as the proposition focuses on the climate-related channel.

Step 1: Expected-loss effect. By definition, the physical one-period default intensity satisfies

$$\lambda_t^{g,\mathbb{P}} = \Lambda^g(X_t, B_t, G_t).$$

Under the monotonicity condition $\partial\Lambda^g/\partial G < 0$ in (8), holding (X_t, B_t) fixed yields

$$\frac{\partial\lambda_t^{g,\mathbb{P}}}{\partial G_t} = \frac{\partial\Lambda^g(X_t, B_t, G_t)}{\partial G_t} < 0. \quad (\text{A.16})$$

Step 2: Risk-premium effect—market price of climate risk. Recall the climate innovation $\tilde{X}_{t+1} := X_{t+1} - \mathbb{E}_t[X_{t+1}]$. Under (1), we have $\tilde{X}_{t+1} = \sigma(G_t)\varepsilon_{t+1}$ and hence

$$\text{Var}_t(\tilde{X}_{t+1}) = \sigma(G_t)^2. \quad (\text{A.17})$$

We use the standard SDF normalization from Appendix A.1,

$$\lambda_X(G_t) := -\frac{\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1})}{\mathbb{E}_t[M_{t+1}]}. \quad (\text{A.18})$$

Assumption 1 (Local linear SDF in the climate innovation) *There exists a $\mathcal{F}_t$ -measurable coefficient $m_X > 0$ and a residual ν_{t+1} such that*

$$M_{t+1} = \bar{M}_t + m_X \tilde{X}_{t+1} + \nu_{t+1}, \quad \text{Cov}_t(\nu_{t+1}, \tilde{X}_{t+1}) = 0. \quad (\text{A.19})$$

Assumption 1 is a local projection that is consistent with (5), i.e., $\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1}) > 0$ implies $m_X > 0$. Under (A.19) and (A.17),

$$\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1}) = m_X \text{Var}_t(\tilde{X}_{t+1}) = m_X \sigma(G_t)^2. \quad (\text{A.20})$$

Substituting (A.20) into (A.18) gives

$$\lambda_X(G_t) = -\frac{m_X}{\mathbb{E}_t[M_{t+1}]} \sigma(G_t)^2 < 0. \quad (\text{A.21})$$

Because $\sigma'(G) < 0$ in (2), differentiating yields

$$\frac{\partial \lambda_X(G_t)}{\partial G_t} = -\frac{m_X}{\mathbb{E}_t[M_{t+1}]} \cdot 2\sigma(G_t)\sigma'(G_t) > 0, \quad (\text{A.22})$$

i.e., the market price of climate risk becomes *less negative* (closer to zero) as G_t rises, reflecting reduced climate volatility.

Step 3: Risk-premium effect—sovereign exposure to climate risk. Appendix A.1 defines (up to the first-order price-to-yield scaling)

$$\beta_X^g(G_t) = -(1 + r_t^f)b_X^g(G_t), \quad b_X^g(G_t) := \frac{\text{Cov}_t(D_{t+1}^g, \tilde{X}_{t+1})}{\text{Var}_t(\tilde{X}_{t+1})}. \quad (\text{A.23})$$

Under the maintained structure of the model, worse climate realizations increase default risk, so $\text{Cov}_t(D_{t+1}^g, \tilde{X}_{t+1}) \geq 0$, which implies $b_X^g(G_t) \geq 0$ and therefore

$$\beta_X^g(G_t) \leq 0. \quad (\text{A.24})$$

To sign the comparative statics in G_t , we impose a sufficient regularity condition capturing “adaptation reduces the climate sensitivity of default”:

Assumption 2 (Adaptation weakly lowers default sensitivity to climate innovations)

Holding (X_t, B_t) fixed, $b_X^g(G)$ is weakly decreasing in G :

$$\frac{\partial b_X^g(G)}{\partial G} \leq 0. \quad (\text{A.25})$$

Assumption 2 is mild in our setup because G reduces both the persistence and

volatility of climate shocks, which attenuates the impact of unexpected climate realizations on the government's repayment capacity. Under (A.25), we have

$$\frac{\partial \beta_X^g(G_t)}{\partial G_t} = -(1 + r_t^f) \frac{\partial b_X^g(G_t)}{\partial G_t} \geq 0. \quad (\text{A.26})$$

Step 4: Putting pieces together. Differentiating the climate risk premium term in (12) yields

$$\frac{\partial}{\partial G_t} \left(\beta_X^g(G_t) \lambda_X(G_t) \right) = \underbrace{\frac{\partial \beta_X^g(G_t)}{\partial G_t}}_{\geq 0} \underbrace{\lambda_X(G_t)}_{< 0} + \underbrace{\beta_X^g(G_t)}_{\leq 0} \underbrace{\frac{\partial \lambda_X(G_t)}{\partial G_t}}_{> 0} \leq 0. \quad (\text{A.27})$$

Combining (A.16) and (A.27) with (12) establishes

$$\frac{\partial s_t^g}{\partial G_t} < 0,$$

as stated in Proposition 2.

A.3. Proof of Proposition 3: Decomposition of corporate credit spreads

Fix a firm i . The one-period defaultable corporate bond price is

$$P_{i,t}^c = \mathbb{E}_t [M_{t+1} (1 - D_{i,t+1}^c)], \quad (\text{A.28})$$

and the default-free price is $P_t^f = \mathbb{E}_t [M_{t+1}]$ with $1 + r_t^f = 1/P_t^f$.

Define the physical one-period default probability (intensity)

$$\lambda_{i,t}^{c,\mathbb{P}} := \mathbb{E}_t[D_{i,t+1}^c]. \quad (\text{A.29})$$

Step 1: Price wedge relative to the risk-free bond. Rewrite (A.28) as

$$\begin{aligned} P_{i,t}^c &= \mathbb{E}_t[M_{t+1}] - \mathbb{E}_t[M_{t+1}D_{i,t+1}^c] \\ &= P_t^f - \left(\mathbb{E}_t[M_{t+1}] \mathbb{E}_t[D_{i,t+1}^c] + \text{Cov}_t(M_{t+1}, D_{i,t+1}^c) \right) \\ &= P_t^f (1 - \lambda_{i,t}^{c,\mathbb{P}}) - \text{Cov}_t(M_{t+1}, D_{i,t+1}^c). \end{aligned} \quad (\text{A.30})$$

Hence,

$$\frac{P_{i,t}^c}{P_t^f} = 1 - \lambda_{i,t}^{c,\mathbb{P}} - \frac{\text{Cov}_t(M_{t+1}, D_{i,t+1}^c)}{P_t^f}. \quad (\text{A.31})$$

Step 2: Linearize the corporate credit spread. Define $1 + r_{i,t}^c := 1/P_{i,t}^c$ and $s_{i,t}^c := r_{i,t}^c - r_t^f$. As in Appendix A.1, let

$$x_{i,t} := \lambda_{i,t}^{c,\mathbb{P}} + \frac{\text{Cov}_t(M_{t+1}, D_{i,t+1}^c)}{P_t^f}, \quad \frac{P_{i,t}^c}{P_t^f} = 1 - x_{i,t}.$$

For small $x_{i,t}$, $(1 - x)^{-1} \approx 1 + x$ implies

$$s_{i,t}^c \approx \lambda_{i,t}^{c,\mathbb{P}} + \frac{\text{Cov}_t(M_{t+1}, D_{i,t+1}^c)}{P_t^f}. \quad (\text{A.32})$$

Step 3: Factor decomposition of the covariance term. Recall $\tilde{X}_{t+1} := X_{t+1} - \mathbb{E}_t[X_{t+1}]$ denote the climate innovation. Project the default indicator on the span of

$\tilde{X}_{t+1}$ and other systematic innovations $\tilde{Z}_{t+1}$:

$$D_{i,t+1}^c = \lambda_{i,t}^{c,\mathbb{P}} + b_{i,X}^c(G_t)\tilde{X}_{t+1} + (b_{i,Z}^c)^\top \tilde{Z}_{t+1} + \varepsilon_{i,t+1}^c, \quad \text{Cov}_t(\varepsilon_{i,t+1}^c, \tilde{X}_{t+1}) = 0. \quad (\text{A.33})$$

Then,

$$\text{Cov}_t(M_{t+1}, D_{i,t+1}^c) = b_{i,X}^c(G_t)\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1}) + (b_{i,Z}^c)^\top \text{Cov}_t(M_{t+1}, \tilde{Z}_{t+1}) + \text{Cov}_t(M_{t+1}, \varepsilon_{i,t+1}^c). \quad (\text{A.34})$$

Define the (conditional) market price of climate risk (same as in Appendix A.1)

$$\lambda_X(G_t) := -\frac{\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1})}{\mathbb{E}_t[M_{t+1}]} = -\frac{\text{Cov}_t(M_{t+1}, \tilde{X}_{t+1})}{P_t^f}, \quad (\text{A.35})$$

and define the corporate bond's climate beta (under the same first-order scaling as in Appendix A.1)

$$\beta_{i,X}^c(G_t) := -(1 + r_t^f) b_{i,X}^c(G_t). \quad (\text{A.36})$$

Substituting (A.34) into (A.32) and using (A.35)–(A.36) yields

$$s_{i,t}^c \approx \lambda_{i,t}^{c,\mathbb{P}} + \beta_{i,X}^c(G_t)\lambda_X(G_t) + (\text{other risk premia}). \quad (\text{A.37})$$

This corresponds to equation (19) in the main text.

A.4. Proof of Proposition 4: Climate adaptation and corporate financing costs

Fix (X_t, Z_i) and consider the local linearization underlying Proposition 3. From (A.37),

$$s_{i,t}^c \approx \lambda_{i,t}^{c,\mathbb{P}} + \beta_{i,X}^c(G_t)\lambda_X(G_t) + (\text{other risk premia}). \quad (\text{A.38})$$

Step 1: Expected-loss effect. By the reduced-form default intensity in the main text,

$$\lambda_{i,t}^{c,\mathbb{P}} = \Lambda^c(X_t, G_t; Z_i).$$

Under $\partial\Lambda^c/\partial G < 0$ in (16), holding (X_t, Z_i) fixed,

$$\frac{\partial\lambda_{i,t}^{c,\mathbb{P}}}{\partial G_t} < 0. \quad (\text{A.39})$$

Step 2: Market price of climate risk moves toward zero with adaptation.

From the climate innovation $\tilde{X}_{t+1} = \sigma(G_t)\varepsilon_{t+1}$ and $\sigma'(G_t) < 0$, Appendix A.2 implies $\lambda_X(G_t) < 0$ and $\lambda'_X(G_t) > 0$ (i.e., λ_X becomes less negative):

$$\lambda_X(G_t) < 0, \quad \frac{\partial\lambda_X(G_t)}{\partial G_t} > 0. \quad (\text{A.40})$$

Step 3: Corporate climate beta becomes less negative with adaptation

(sufficient condition). Recall $\beta_{i,X}^c(G_t) = -(1 + r_t^f)b_{i,X}^c(G_t)$ where $b_{i,X}^c$ is the projection of $D_{i,t+1}^c$ on $\tilde{X}_{t+1}$. Because corporate default risk increases with adverse

climate realizations, we have $b_{i,X}^c(G_t) \geq 0$ and hence

$$\beta_{i,X}^c(G_t) \leq 0. \quad (\text{A.41})$$

We impose a natural exposure-reduction condition: holding (X_t, Z_i) fixed,

$$\frac{\partial b_{i,X}^c(G)}{\partial G} \leq 0 \quad \implies \quad \frac{\partial \beta_{i,X}^c(G)}{\partial G} \geq 0. \quad (\text{A.42})$$

(That is, adaptation weakly lowers the sensitivity of corporate default risk to climate innovations.)

Step 4: Risk-premium effect is weakly negative. Differentiate the climate risk-premium term:

$$\frac{\partial}{\partial G_t} (\beta_{i,X}^c(G_t) \lambda_X(G_t)) = \underbrace{\beta_{i,X}^{c'}(G_t)}_{\geq 0} \underbrace{\lambda_X(G_t)}_{< 0} + \underbrace{\beta_{i,X}^c(G_t)}_{\leq 0} \underbrace{\lambda_X'(G_t)}_{> 0} \leq 0. \quad (\text{A.43})$$

Assuming the remaining “other risk premia” in (A.38) are locally insensitive to G_t conditional on (X_t, Z_i) (or weakly decreasing in G_t), (A.39) and (A.43) imply

$$\frac{\partial s_{i,t}^c}{\partial G_t} < 0. \quad (\text{A.44})$$

This establishes the Proposition 4.

A.5. Proof of Corollary 2: Cross-sectional predictions for corporate spreads

Fix time t and consider a marginal increase in adaptation investment G_t . Under the same local approximation as in Proposition 3 and Proposition 4, the corporate credit spread satisfies (19)

$$s_{i,t}^c \approx \lambda_{i,t}^{c,\mathbb{P}} + \beta_{i,X}^c(G_t)\lambda_X(G_t) + (\text{other risk premia}).$$

We show that, for a given change $\Delta G > 0$, the decline in $s_{i,t}^c$ is larger for firms with larger θ_i and larger η_i in (14).

Step 1: A structural representation of default probability. To make the dependence on (θ_i, η_i) explicit, consider the standard threshold form for one-period default:

$$D_{i,t+1}^c = \mathbf{1}\{CF_{i,t+1} < \bar{C}_i\},$$

where $\bar{C}_i$ is a firm-specific obligation threshold. Using the cash-flow specification (14),

$$CF_{i,t+1} = \bar{C}F_i - \theta_i X_{t+1} - \eta_i \lambda_t^g + \epsilon_{i,t+1}.$$

Assume $\epsilon_{i,t+1}$ has a continuously differentiable cdf F_ϵ with density $f_\epsilon > 0$ in the relevant region. Conditioning on $\mathcal{F}_t$ yields the physical default probability

$$\begin{aligned} \lambda_{i,t}^{c,\mathbb{P}} &= \mathbb{E}_t[D_{i,t+1}^c] = \Pr_t(CF_{i,t+1} < \bar{C}_i) \\ &= F_\epsilon\left(\bar{C}_i - \bar{C}F_i + \theta_i \mathbb{E}_t[X_{t+1}] + \eta_i \lambda_t^g\right). \end{aligned} \quad (\text{A.45})$$

Step 2: How adaptation affects the two arguments in (A.45). First, from the climate dynamics (1),

$$\mathbb{E}_t[X_{t+1}] = (1 - \phi(G_t))X_t.$$

Since X_t is an intensity (nonnegative) and $\phi'(G) > 0$ by (2), we have

$$\frac{\partial \mathbb{E}_t[X_{t+1}]}{\partial G_t} = -\phi'(G_t)X_t < 0. \quad (\text{A.46})$$

Second, by Proposition 2, holding (X_t, B_t) fixed,

$$\frac{\partial \lambda_t^g}{\partial G_t} < 0. \quad (\text{A.47})$$

Step 3: Cross-sectional heterogeneity via the expected-loss channel. Differentiate (A.45) with respect to G_t and apply the chain rule:

$$\frac{\partial \lambda_{i,t}^{c,\mathbb{P}}}{\partial G_t} = f_\epsilon(\cdot) \left[\theta_i \frac{\partial \mathbb{E}_t[X_{t+1}]}{\partial G_t} + \eta_i \frac{\partial \lambda_t^g}{\partial G_t} \right]. \quad (\text{A.48})$$

Using (A.46)–(A.47), both derivatives inside the bracket are negative. Therefore,

$$\frac{\partial \lambda_{i,t}^{c,\mathbb{P}}}{\partial G_t} < 0, \quad \frac{\partial}{\partial \theta_i} \left(\frac{\partial \lambda_{i,t}^{c,\mathbb{P}}}{\partial G_t} \right) = f_\epsilon(\cdot) \frac{\partial \mathbb{E}_t[X_{t+1}]}{\partial G_t} < 0, \quad \frac{\partial}{\partial \eta_i} \left(\frac{\partial \lambda_{i,t}^{c,\mathbb{P}}}{\partial G_t} \right) = f_\epsilon(\cdot) \frac{\partial \lambda_t^g}{\partial G_t} < 0.$$

Hence, the reduction in expected-loss (and thus in spreads) is amplified by θ_i and η_i .

Step 4: Adding the climate risk-premium term does not overturn the heterogeneity. From Proposition 4 / Appendix proof, the climate risk-premium component $\beta_{i,X}^c(G_t)\lambda_X(G_t)$ is weakly decreasing in G_t (i.e., its G -derivative is ≤ 0). Moreover, in the cash-flow equation (14), the firm-specific exposure to X_{t+1} is scaled by θ_i , so the firm's climate-related default exposure (and thus $|\beta_{i,X}^c|$) is (weakly) increasing in θ_i , which further strengthens the spread relief for more climate-exposed firms.

Combining Step 3 and Step 4 with the spread decomposition (19), and assuming the remaining “other risk premia” are locally insensitive to G_t (or weakly decreasing) conditional on (X_t, Z_i) , we conclude that for any small $\Delta G > 0$,

$$\Delta s_{i,t}^c \approx \frac{\partial s_{i,t}^c}{\partial G_t} \Delta G$$

is more negative (a larger spread reduction) when θ_i is larger and/or η_i is larger. This establishes Corollary 2.

References

Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, Princeton, NJ, revised edition edition.